Lab 7(A) - Laplacian Filter

In [ ]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

Apply Laplacian Filter on Image to get Sharp Image

In [ ]:

```
# ================ ( Laplacian Filter ) ====================


imgFile = "moon.jpg"

img = cv2.imread(imgFile, 0)

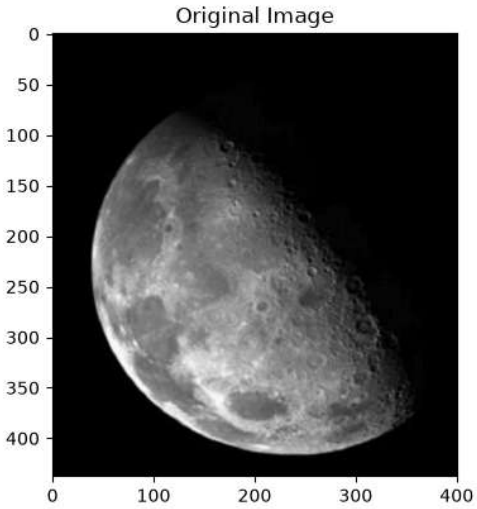

manual_kernel = np.array([
    [0,  1,  0],
    [1, -4,  1],
    [0,  1,  0]
], dtype=np.float32)

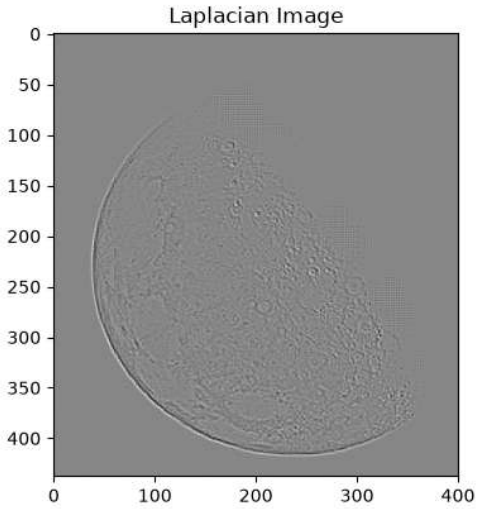

laplacian_manual = cv2.filter2D(img, cv2.CV_64F, manual_kernel)

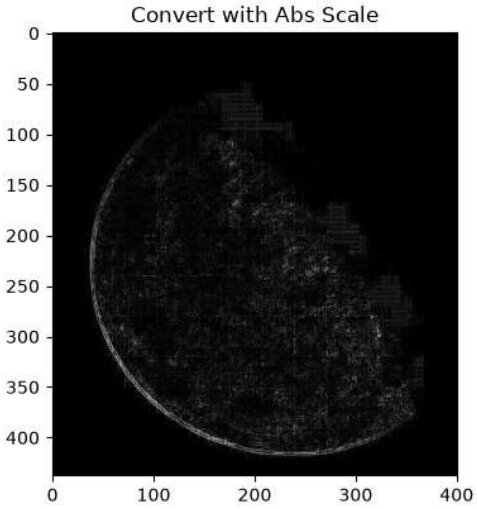
laplacian_abs = cv2.convertScaleAbs(laplacian_manual)

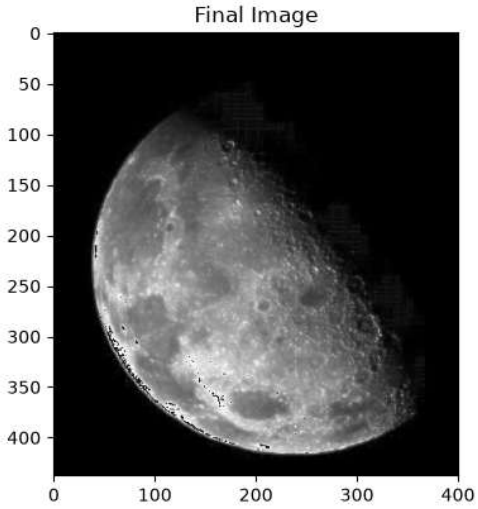
finalImage = img + laplacian_abs

plt.figure(figsize=(12, 10))
plt.subplot(221), plt.imshow(img, cmap='gray'), plt.title('Original Image')
plt.subplot(222), plt.imshow(laplacian_manual, cmap='gray'),
plt.title('Laplacian Image')
plt.subplot(223), plt.imshow(laplacian_abs, cmap='gray'), plt.title('Convert
with Abs Scale')
plt.subplot(224), plt.imshow(finalImage, cmap='gray'), plt.title('Final
Image')
plt.show()
```


Original Image
0
50
100
150
200
250
300
350
400
0
100
200
300
400


Laplacian Image
0
50
100
150
200
250
300
350
400
0
100
200
300
400


Convert with Abs Scale
0
50
100
150
200
250
300
350
400
0
100
200
300
400


Final Image
0
50
100
150
200
250
300
350
400
0
100
200
300
400

In [ ]:

```
blurred = cv2.GaussianBlur(img, (3, 3), 0)


laplacian = cv2.Laplacian(blurred, cv2.CV_64F)

laplacian_8bit = cv2.convertScaleAbs(laplacian)

finalImage2 = img + laplacian_8bit

plt.figure(figsize=(12, 10))
plt.subplot(221), plt.imshow(img, cmap='gray'), plt.title('Original Image')
plt.subplot(222), plt.imshow(laplacian, cmap='gray'), plt.title('Laplacian
Image')
plt.subplot(223), plt.imshow(laplacian_8bit, cmap='gray'),
plt.title('Convert with Abs Scale')
plt.subplot(224), plt.imshow(finalImage2, cmap='gray'), plt.title('Final
Image')
plt.show()
```

